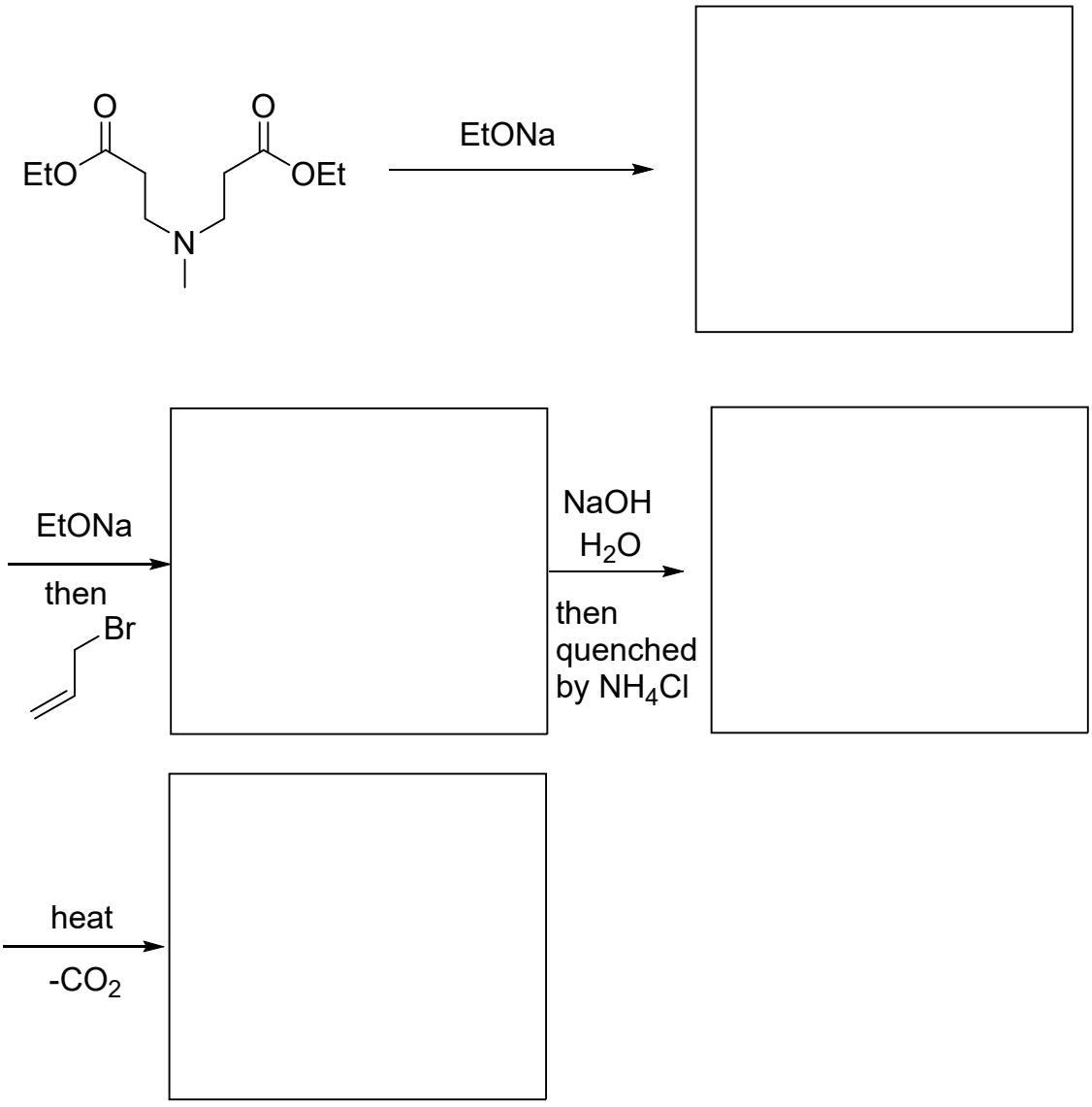
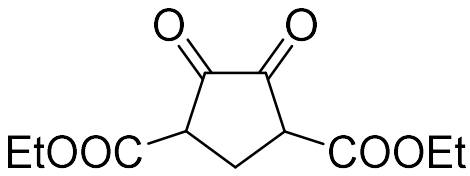
Question 1. Propose the structures of the intermediates and the products



Question 2. Propose the synthesis of the following compound. Use starting materials that contain maximum three consecutive carbon chains (O, N, H atoms are not counted, and reagents that do not end up in the target molecules are not considered).



Question 3. Propose stepwise mechanisms for the following reaction. (Hint: An enamine is formed at first.)

